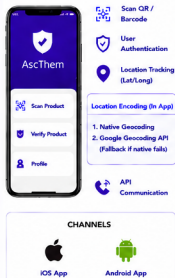
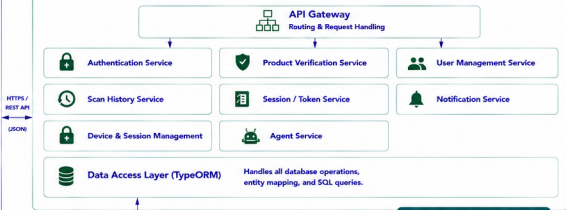


AscThem - High Level Architecture Diagram

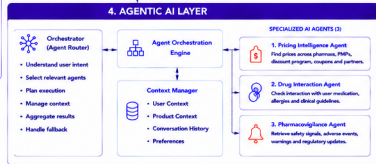
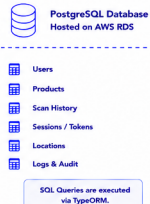
1. CLIENT (MOBILE APP)



2. BACKEND (APPLICATION SERVER) – Deployed on AWS (Lightsail / Containers)



3. DATABASE (POSTGRESQL) on AWS RDS



5. DATA & KNOWLEDGE SOURCES

FDA SOURCES (Authoritative)

- Drugs@FDA
- Approved drug information
- NDC Directory
- Product & package information
- Orange Book
- Approved products & therapeutic equivalence
- FDALabel
- Full text drug labeling information (Warnings, etc.)
- FAERS
- Adverse Event Reporting System
- Safety Communications
- Drug Safety & Availability
- Recalls / Market Withdrawals
- Recall & safety alerts

EXTERNAL SOURCES (Non-FDA)

- Pharmacy Pricing APIs
- Retail & online pharmacies
- Coupon & Savings Providers
- Manufacturer & thirdparty offers
- Review & Rating Platforms
- Customer reviews & ratings
- Drug Interaction Databases (e.g., RxNorm, Micromedex, etc.)

6. EXTERNAL SERVICES (INTEGRATIONS)



7. LLM & AI SERVICES LAYER



8. INFRASTRUCTURE LAYER (AWS)



10. SECURITY & GOVERNANCE (CROSS-CUTTING)



11. ORCHESTRATION FLOW (HIGH LEVEL)

1. User scans product → Basic info displayed from barcode & label.
2. User clicks Verify → Backend validates via services and DB.
3. Orchestrator understands context and routes to relevant agents.
4. Agents run in parallel (where possible) and return structured results.
5. Orchestrator aggregates, reasons and generates final intelligent response.
6. User receives clear, actionable insights and recommendations.

LEGEND

- Request / Response
- - - - - Data / Context Flow
- - - - - Event / Async Flow
- ↔ Bi-directional Flow